Fourier Transforms

Consider the heat equation on a compact interval:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \tag{1}$$

where u is a solution. If u represents a rod that is at time $t = 0$ half hot and half cold and is touching in the middle, then we let time move forward, the temperature will dampen out due to the 2nd term. The shape goes from a step function, to a function that looks like $\tan^{-1}(x)$, and as time $\to \infty$ it converges to a constant function representing the average temperature. These are all functions with distinctly different behaviours.

Now, assume that the heat of the rod at $t = 0$ has the shape of $\sin(x)$ or $\cos(x)$. Then as we let time propagate these functions dampen uniformly, notably *they are always the same function*, simply scaled. This is essentially the behaviour of *eigenvectors*, namely as time moves forward we are scaling by a factor. To have eigenvectors, we need a linear transformation between vector spaces. Let S_t be the *time evolution operator* that takes in a state f_{t_0} (a function on the domain that is a t_0-slice of a solution $u(x, t_0)$) and outputs f_{t_0+t}, in particular $S_t : \mathscr{P}(X) \to \mathscr{P}(X)$ where $\mathscr{P}(X)$ is the state space over X and S_t is the evolve by t operator:

$$S_t(u(-, t_0)) = u(-, t_0 + t)$$

Or, viewed as an (abelian) monoid action[1]:

$$S : [0, \infty) \times \mathscr{P}(X) \to \mathscr{P}(X) \qquad t \cdot f_{t_0} = f_{t_0+t}$$

Note that it isn't "obvious" what type of space $\mathscr{P}(X)$ is, we will harmlessly assume that this is the Hilbert spaces; see [3] for the justification of this assumption. At minimum it is certainly the case that any (smooth) L^2-integrable function is a valid starting state that can then evolve according to the heat equation. If the reader can accept that there is a reasonable sense in which integrable function can be differentiated as shall be demonstrated [2], the reader can let $\mathscr{P}(X) = L^2([0, 1])$.

Now, certainly S_t is linear as the heat equation is linear and homogeneous[2]. We can thus ask if it has eigenvectors, that is function $f_r \in \mathscr{P}(X)$ where

$$S_t f_r(x) = \lambda_t f_r(x)$$

If we let t vary, we can define a function $\lambda(t)$ dependent on the time. Given a compact domain $X = [a, b]$, by direct computations we can check that $\sin(kx)$ is an eigenvector, where this is indeed a state as it is a t-slice of:

$$u(x, t) = e^{-k^2 t} \sin(kx)$$

[1] On the full real line the forward heat equation is still well-posed in $L^2(\mathbb{R})$. What fails is uniqueness across *all* solutions (Tychonoff exhibited a nonzero solution with zero initial data), so a growth condition is needed to pin one down; the *backward* equation is the genuinely ill-posed one. See [3]

[2] It is also continuous and normal, but for that we would need to choose whether we consider $\mathscr{P}$ as a Banach space or a distribution space; intuitively the heat equation dissipates energy so $\|S_t f\| \leq \|f\|$

That is, for every t:

$$S_t(\sin(kx)) = e^{-k^2 t}\sin(kx)$$

With the assumption that $\mathscr{P}$ is a Hilbert space, if these eigenfunctions form an orthonormal basis, we can decompose a functions into a "weighted sum" of these well-behaved functions with respect to time. For the rest of this section we shall take for granted $\{\sin(kx)\}_{k\in\mathbb{N}}$ form an orthonormal basis of $\mathscr{P}(X)$ diagonalising S_t (proven in [2]) after appropriately normalized depending on the domain.

It must be pointed out that the same argument works for $\cos(kx)$:

$$S_t(\cos(kx)) = e^{-k^2 t}\cos(kx)$$

and these *also* form an orthonormal basis that *also* diagonalize S_t. These two sets certainly cannot be combined since the result is no longer orthogonal:

$$\int_0^1 \sin(\pi x)\cos(2\pi x)dx = \frac{-2}{3\pi} \neq 0$$

Thus, the question shifts into one about finding a *natural* eigenbasis for S_t. A-priori, it is not necessarily the case that these are the only eigenbasis. What we need is for the eigenbasis to "respect" some inherent geometry of the space. Phrased differently, the eigenbasis needs to be *invariant* with respect to some operator(s), where the operators represent some symmetry of the space. To get a better grasp of invariance of operators, we may start looking at other PDEs and considering diagonalizing the operator S_t with respect to the solutions of those PDEs.

For example, if we solve the Heat equation on a sphere:

$$\frac{\partial u}{\partial t} = \Delta_{\text{sphere}} u$$

the operator can still be diagonalized (see [3]). What we shall want is for an eigenbasis that respects the spherical geometry, namely for it to be invariant to an $SO(3)$ action. Such eigenfunctions do exist, they are the *Spherical Harmonics* $Y_l^m(\theta,\varphi)$. No individual Y_l^m is fixed by rotation (rotating one returns a combination of the $Y_l^{m'}$), but the $(2l+1)$-dimensional span of $\{Y_l^m\}_{m=-l}^{l}$ is rotation-invariant, and the Laplacian itself commutes with rotations[3]

$$\Delta(f\circ R) = (\Delta f)\circ R$$

To give another example that is famous in physics, consider Quantum Harmonic Oscillator

$$i\frac{\partial \psi}{\partial t} = -\frac{\partial^2\psi}{\partial x^2} + x^2\psi$$

This PDE is diagonalizable (see [3]), but it is *not* diagonalizable by $e^{2\pi i k x}$. Instead it was shown to be diagonalized (uniquely) by the *Hermite Functions* (a Gaussian multiplied by a polynomial)

$$\varphi_n(x) = H_n(x)e^{-x^2/2}$$

By trying to diagonalize with $e^{2\pi ikx}$, the x^2 multiplication term turns into a Laplacian in k-space $(-\partial_k^2)$, so the result will effectively just swap the PDE for an identical PDE in frequency space

[3] important technicality: $SO(3)$ is not abelian, hence it cannot be that *all* rotation operators simultaneously commute. The best solution is to pick a "maximal set" of commuting symmetries: usually the Total Angular Momentum (L^2, representing the amount of rotation invariance) and Azimuthal Symmetry (L_z, rotation around the z-axis).

without solving it. In physics, this fact shall be the key to showing that it is in fact *invariant* to the Fourier transform, creating fascinating link between abstract harmonic analysis and the concrete geometry of quantum mechanics, a connection to be explored in another book.

So what is the invariance we want to impose for the operator S_t with respect to the heat equation? In the example of the sphere, the group was $SO(3)$. A compact interval has no *translation* invariance, simply because it is not a group, though it does carry a reflection symmetry, which we shall put to work shortly. However, by gluing the end points of the interval we get a 1-*torus* (a circle)[4]. We now have the rotation group operation on $\mathbb{T}^1$, or equivalently the translation operation on $\mathbb{R}/\mathbb{Z}$[5]. This does shift the geometry and hence the inherent dynamics of our PDEs; the problem of solving a PDE on a compact interval will be a subset of solving PDEs on a torus as shall be addressed shortly. Let us first simply consider S_t on $L^2(\mathbb{T}^1)$ and see how it interacts with the translation operator. Since we are working on the torus, we now have that $\sin(2\pi kx)$ and $\cos(2\pi lx)$ *are* orthogonal (where the 2π-scaling was to ensure a full period of each lies within the torus), and $\{\sin(kx), \cos(lx)\}_{k,l\in\mathbb{N}}$ is an orthonormal basis (after appropriate scaling and stretching). Now, note that sin and cos basis are *not* translation invariant, for example:

$$\tau_a(\sin(kx)) = \cos(ka)\sin(kx) + \sin(ka)\cos(kx)$$

and so they are not eigenvectors of τ_a. However, they still *block-diagonalize* the translation operator τ_a with blocks of size 2:

$$\begin{pmatrix} \cos(ka) & \sin(ka) \\ -\sin(ka) & \cos(ka) \end{pmatrix}$$

If we look at these blocks, we shall see that they are all rotation matrices by an angle ka; in other words the problem is that over $\mathbb{R}$ we did not have enough eigenvalues. By extending the codomain to $\mathbb{C}$, we can readily verify that:

$$\tau_a(e^{iz}) = e^{ia} \cdot e^{iz}$$

and so $\{e^{ikz}\}_{k\in\mathbb{Z}}$ is our desired eigenbasis for τ_a. Now, do these translations of these functions commute with S_t, namely:

$$S_t(\tau_a(f)) = \tau_a(S_t(f)) \qquad \forall a \in \mathbb{T}^1$$

Indeed:

$$\begin{aligned} \tau_a(S_t(e^{ikx})) &= \tau_a(e^{-k^2t}e^{ikx}) \\ &= e^{-k^2t}e^{ik(x+a)} \\ &= e^{-k^2t}e^{ika}e^{ikx} \\ &= e^{ika}S_t(e^{ikx}) \\ &= S_t(e^{ika}e^{ikx}) && S_t \text{ is linear} \\ &= S_t(\tau_a(e^{ikx})) \end{aligned}$$

What this means physically is that the heat equation dissipates the same way no matter where you are on a ringed-shape rod. If the rod was half wood and half metal, then this translation invariance shall fail.

[4] More generally, any rectangular domain is homeomorphic to $[0,1]^n$ which then can be glued to become $\mathbb{T}^n = (S^1)^n$, and so this same translation structure appears in higher dimensions

[5] The domain stays $\mathbb{R}/\mathbb{Z}$; it is the *codomain* of our functions that we shall shortly extend to $\mathbb{C}$

Now, notice that as $\tau_a \circ S_t = S_t \circ \tau_a$ for all $a \in \mathbb{T}^1$ (or in commutator bracket notation $[\tau_a, S_t] = 0$ for all a, t), any simultaneous eigenvector of *every* τ_a is also an eigenvector of S_t. The converse fails, and instructively: S_t takes the same value $e^{-k^2 t}$ on e^{ikx} and on e^{-ikx}, so its eigenspaces are two-dimensional, and $\sin(kx)$ is an eigenvector of S_t while, as we just saw, not being one of τ_a. Asking for translation symmetry is exactly what breaks that degeneracy. The eigenvectors of S_t are in general hard to compute, but the simultaneous eigenvectors of the τ_a are *precisely* the functions e^{ikx}.

Boundary Problem

The immediate point that is still pending and must clarified when switching to $\mathbb{T}^1$ is that this shifts the dynamics of any PDE we are solving, as now points at the boundary are affected by their neighbourhood. However, for PDE's that will respect the group structure of $\mathbb{T}^1$, the effects at the boundary can be "cancelled out" in two separate ways that shall recover the original two basis $\{\sin(kx)\}_{k\geq 1}$ and $\{\cos(kx)\}_{k\geq 0}$ we saw before (again, appropriately normalized). The key insight is that the interval $[0, 1]$ can be viewed as a *sub-domain* of the torus $\mathbb{T}^1$ (identified with $[-1, 1]$) subject to a symmetry constraint. While the torus possesses full translation invariance, the interval possesses *reflection invariance*. By imposing a symmetry on the torus, we create "virtual boundaries" at the fixed points of the reflection ($x = 0$ and $x = 1$). This gives rise to two distinct orthonormal bases for $L^2([0, 1])$, each corresponding to a different physical boundary condition:

- *Odd Symmetry (Dirichlet Conditions):* Consider the subspace of functions on $\mathbb{T}^1$ that are *odd*, satisfying $u(-x) = -u(x)$. At the fixed point $x = 0$, this symmetry forces $u(0) = -u(0)$, implying $u(0) = 0$. Physically, the heat flowing from the right is perfectly cancelled by the "negative heat" flowing from the left, creating a virtual cold wall. The eigenfunctions respecting this symmetry are precisely the sine functions:

  $$\sin(n\pi x) = \frac{e^{in\pi x} - e^{-in\pi x}}{2i}$$

  Restricting these functions to $[0, 1]$ yields the *Sine Basis*, which diagonalizes the Laplacian subject to $u(0) = u(1) = 0$.

- *Even Symmetry (Neumann Conditions):* Consider the subspace of functions on $\mathbb{T}^1$ that are *even*, satisfying $u(-x) = u(x)$. At the fixed point, the derivative must be odd, forcing $u'(0) = 0$. Physically, this represents an insulated boundary where heat reflects back without loss. The eigenfunctions respecting this symmetry are the cosine functions:

  $$\cos(n\pi x) = \frac{e^{in\pi x} + e^{-in\pi x}}{2}$$

  Restricting these to $[0, 1]$ yields the *Cosine Basis*, which diagonalizes the Laplacian subject to $u'(0) = u'(1) = 0$.

This leads to a result in Harmonic Analysis known as *half-range expansions*: while sines and cosines *together* form a basis for the full torus, *either* set alone forms a complete orthonormal basis for the interval $[0, 1]$.

Decoupling PDE

To understand the consequences better, take a solution $u : \mathbb{T}^1 \times [0,\infty) \to \mathbb{C}$. Then, think of each (space) neighborhood in the usual heat equation as influencing its infinitesimal neighbourhood, namely the $\frac{\partial^2 u}{\partial x^2}$ term determines how each neighbouring point affects each other as time propagates. We can think of this as being uncountably many dependent equations.

Therefore, by applying the Fourier transform, we are changing the basis of our state space from the position "basis" $\{\delta_x\}_{x\in\mathbb{T}^1}$ (distributional rather than an honest basis, since no δ_x lies in L^2) to the genuine eigenbasis of the spatial operator $\{e^{ikx}\}_{k\in\mathbb{Z}}$.

By taking the Fourier transform, i.e. transforming to the translation-invariant orthonormal basis, we change a solution to:

$$u(x,t) : \mathbb{T}^1 \times [0,\infty) \to \mathbb{C} \qquad \xmapsto{\mathcal{F}_x} \qquad \hat{u}(k,t) : \mathbb{Z} \times [0,\infty) \to \mathbb{C}$$

where the $\mathbb{Z}$ is the index for the eigenbasis $\{f_k\}_{k\in\mathbb{Z}}$. Then if $\hat{u}_k$ is the function whose input is the index of the eigenvector and output is its coefficient, we see that as $\hat{u}_k$ propagates through time, $\hat{u}_k(t)$ is only dependent on the original function up to scaling. In fact we shall show that the heat equation PDE decomposes into the equations:

$$\frac{d\hat{u}_k}{dt} = -k^2\hat{u}_k \qquad \forall k \in \mathbb{Z}$$

where now we have *countably many decoupled* equations.

Non-Compact Domain

Notice that $\sin, \cos, e^{ikx}$ are not square-integrable on $\mathbb{R}^n$ (resp. $\mathbb{C}^n$), hence these cannot form an orthonormal basis on $L^2(\mathbb{R}^n)$. Thus, the initial foray into Fourier will focus on functions with domain $\mathbb{T}^n$. In [5] we shall show how to generalize the tools to the non-compact case, more specifically, the locally compact case.

Digression to Lie Theory

We have established that the Fourier basis functions e^{ikx} are the eigenfunctions of the "global" spatial translation operator τ_a. However, the reader likely recognizes these functions primarily as the solutions to the differential equation $f' = \lambda f$. This connection between the *global* translation τ_a and the *local* derivative $D = \frac{d}{dx}$ is a central theme of Lie Theory.

In [2], it shall be shown that it makes sense to infinitely differentiate integrable functions. For now, we rely on the intuition that any continuous function can be approximated by a smooth function (and locally by a polynomial).

Consider the Taylor expansion of a function f around a point x:

$$f(z) = f(x) + f'(x)(z-x) + \frac{f''(x)}{2!}(z-x)^2 + \dots$$

If we wish to evaluate f at a shifted point $x+y$, the Taylor series gives:

$$f(x+y) = f(x) + yf'(x) + \frac{y^2}{2!}f''(x) + \dots$$

Notice that we can re-write this using the differentiation operator $D = \frac{d}{dx}$:

$$f(x+y) = \left(I + yD + \frac{y^2D^2}{2!} + \dots\right) f(x)$$

Recognizing the series inside the parenthesis as the definition of the operator exponential, we arrive at the compact operator identity:

$$\tau_y = e^{yD}$$

This equation states that *differentiation generates translation.* Taken literally the Taylor step asks for f to be *analytic* (a smooth function's Taylor series need not converge back to it, as e^{-1/x^2} shows), which the basis functions e^{ikx} certainly are. The identity holds far more widely in the semigroup sense, with D the generator of the translation group; see [2].

Now, consider the spectral consequences. If an operator A has an eigenvector v with eigenvalue λ ($Av = \lambda v$), then the operator exponential e^A acts on v as:

$$e^A v = \left(I + A + \frac{A^2}{2!} + \dots\right) v = \left(1 + \lambda + \frac{\lambda^2}{2!} + \dots\right) v = e^\lambda v$$

Applying this to our spatial operators: since $f(x) = e^{ikx}$ is an eigenfunction of the derivative D with eigenvalue ik (i.e., $De^{ikx} = ike^{ikx}$), it must automatically be an eigenfunction of the translation operator $\tau_y = e^{yD}$ with eigenvalue e^{iyk}.

This confirms our earlier finding: the Fourier basis e^{ikx} simultaneously diagonalizes the derivative (and by extension the Laplacian $\Delta = D^2$) and the translation group.

Theorem 0.1: Eigenvectors Under the Operator Exponential

Any eigenvector with eigenvalue λ of an operator A is an eigenvector of the operator e^A with eigenvalue e^λ.

This is the easy half of the *spectral mapping theorem* $\sigma(f(A)) = f(\sigma(A))$; for exponentials of unbounded generators the full statement can fail, so it is worth keeping the two apart.

> ***Proof*** **:**
> Follows from the power series definition of the operator exponential shown above.

To see this explicitly, observe the action of these operators on the basis function $f_k(x) = e^{ikx}$:

- *Derivative:* $Df_k = (ik)f_k$ (Eigenvalue ik)
- *Laplacian:* $\Delta f_k = D^2 f_k = (ik)^2 f_k = -k^2 f_k$ (Eigenvalue $-k^2$)
- *Translation:* $\tau_a f_k = e^{aD} f_k = e^{a(ik)} f_k$ (Eigenvalue e^{ika})

In the language of representation theory, the differentiation operator is an element X of the *Lie algebra* $\mathfrak{g}$, and the translation operator is the *Lie group* element e^X. The Fourier basis provides the unique coordinate system where the group action and its generator are both simple scalings.

Generalizing using Representation Theory

In the previous section, our motivation was to solve a specific time-dependent PDE (the heat equation). We discovered that the key to solving it lay not in the time evolution S_t itself, but in the underlying symmetry of the spatial domain (translation invariance). The basis functions e^{ikx} worked because they diagonalized the translation operator τ_a.

We now abstract this principle. We no longer need a specific "time" equation; instead, we look for the basis that diagonalizes the action of the symmetry group itself. By finding the irreducible representations of the symmetry group, we construct a "universal" eigenbasis that will automatically diagonalize *any* linear operator respecting that symmetry (be it the Laplacian, the Heat Operator, or others).

To make this concrete, let us recall the representation theory of finite groups. If G is a finite group, we can represent G on a vector space V (over a field k) by taking a homomorphism $\rho : G \to \mathrm{GL}(V)$.

$$G \curvearrowright V \qquad \rho(g)v = g \cdot v$$

As we are not interested in any arithmetic limitations, we take $k = \mathbb{C}$. Then by Maschke's theorem (see [4, chapter 24]) the induced $\mathbb{C}[G]$-module structure on V is semi-simple, meaning:

$$V \cong_G \bigoplus_i V_i$$

where V_i are irreducible representations. Given the basis for the above isomorphism, any $g \in G$ will give a matrix representation $\rho(g)$ which respects this block-diagonal decomposition:

$$\rho(g) = \begin{bmatrix} \mathbf{M}_1(g) & 0 & 0 & 0 \\ 0 & \mathbf{M}_2(g) & 0 & 0 \\ 0 & 0 & \mathbf{M}_3(g) & 0 \\ 0 & 0 & 0 & \dots \end{bmatrix}$$

where $\mathbf{M}_i(g)$ represents the matrix block acting on the subspace V_i. Crucially, if G is *abelian*, Schur's Lemma implies that *all* irreducible representations are 1-dimensional. Hence, the matrices become scalars:

$$\rho(g) = \begin{bmatrix} \lambda_1(g) & 0 & 0 & 0 \\ 0 & \lambda_2(g) & 0 & 0 \\ 0 & 0 & \lambda_3(g) & 0 \\ 0 & 0 & 0 & \dots \end{bmatrix}$$

As each $\rho(g)$ is representable by a diagonal matrix using the same basis, the operators $\{\rho(g)\}_{g \in G}$ are a commuting family of linear operators (i.e. they are *simultaneously diagonalizable*).

Now, we generalize this to function spaces. While for a finite group the vector space of functions $\mathbb{C}[G]$ is finite-dimensional, equipping it with an inner product structure is essential to define unitarity and orthogonality. A finite group G carries a natural measure, unique up to scale (the counting measure), which we can normalize to be a probability measure (the Haar measure $\mu(g) = 1/|G|$). This allows us to define the Hilbert space $L^2(G)$ with inner product:

$$\langle f, h \rangle = \frac{1}{|G|} \sum_{x \in G} f(x)\overline{h(x)}$$

We define the *Right Regular Representation*[6] of G on this space:

$$G \curvearrowright L^2(G) \qquad (R_g \cdot f)(x) = f(xg)$$

We verify that this action is unitary (and hence can be diagonalized with an orthonormal basis). Since right-multiplication by g is a bijection on the group G (it permutes the elements), the sum over x is identical to the sum over $y = xg$:

$$\begin{aligned}\langle R_g f, R_g h\rangle &= \frac{1}{|G|}\sum_{x\in G} f(xg)\overline{h(xg)} \\ &= \frac{1}{|G|}\sum_{y\in G} f(y)\overline{h(y)} \\ &= \langle f, h\rangle\end{aligned}$$

Thus, the operators R_g are unitary. Since G is abelian, the operators $\{R_g\}_{g\in G}$ form a commuting family of unitary operators. By the spectral theorem for unitary operators, there exists an orthonormal basis $\mathcal{B}$ of $L^2(G)$ that simultaneously diagonalizes the action. If $\varphi \in \mathcal{B}$ is such a basis vector, then:

$$R_g(\varphi) = \chi_\varphi(g)\varphi$$

where $\chi_\varphi(g) \in \mathbb{C}$ is the eigenvalue. Since the representation is a homomorphism ($R_{gh} = R_g R_h$), the eigenvalues must satisfy the multiplicative property $\chi_\varphi(gh) = \chi_\varphi(g)\chi_\varphi(h)$. Moreover, since R_g is unitary, $|\chi_\varphi(g)| = 1$ and hence the codomain is S^1. These homomorphisms $\chi : G \to S^1$ are precisely the *characters* of the group. Thus,

> the group-theoretic eigenbasis for $L^2(G)$ is simply the set of characters!

We shall explore this more in [5]

0.2 Advanced: Operators Commuting with Translations We focused on finding the basis that diagonalizes translations (R_g). But what are the operators that commute with all translations?

$$T \circ R_g = R_g \circ T \qquad \forall g \in G$$

For G finite, every such T is a *Convolution Operator*: there is a $k \in L^1(G)$ (the kernel) with $Tf = f * k$. They are the most general linear maps respecting the symmetry of the space, which is why the Heat Operator and its relatives are convolutions.
Past the finite case the kernel need not be a function. The Laplacian is convolution against a derivative of δ, not against anything in L^1; and on $\mathbb{R}$ the translation-invariant bounded operators on L^2 are exactly the Fourier multipliers with L^∞ symbol, strictly more than convolution against L^1, whose symbols must vanish at infinity.

The Infinite Compact Case

In the case where the domain is infinite (like the torus $\mathbb{T}^n$), we must be more careful. First, let us stick to compact domains so that we generalize the finite case directly. Functions defined on

[6] or, Left Regular Representations $(L_g \cdot f)(x) = f(g^{-1}x)$

a rectangular domain with periodic boundary conditions are equivalent to functions on the torus $\mathbb{T}^n = (S^1)^n$. Our state space is the Hilbert space $L^2(\mathbb{T}^n, \mathbb{C})$.

We want to act by translation by $\mathbb{R}^n$, but since the domain is periodic, this descends to a faithful action of the compact group $\mathbb{R}^n/\mathbb{Z}^n \cong \mathbb{T}^n$. It is easy to check that translation is unitary ($\langle \tau_y f, \tau_y g\rangle = \langle f, g\rangle$) using the translation invariance of the Lebesgue measure.

We now invoke the *Peter-Weyl Theorem*, which is the generalization of Maschke's theorem to compact topological groups (see [1]). It states that the unitary representation of a compact group on $L^2(G)$ decomposes into a Hilbert direct sum of finite-dimensional irreducible representations:

$$L^2(G) \cong_G \widehat{\bigoplus_i} V_i$$

Since $\mathbb{T}^n$ is abelian, all irreducible representations V_i are *one-dimensional*. Being one-dimensional and unitary, they are given by characters $\chi : \mathbb{T}^n \to S^1$.

Thus, the Peter-Weyl theorem guarantees that the characters of $\mathbb{T}^n$ form a complete orthonormal basis for $L^2(\mathbb{T}^n)$. We have recovered our Fourier basis purely from the algebraic structure of the domain, without reference to any specific differential equation!

* The Duality of Points and Spectrum

The statement that the Fourier transform changes the basis from "positions" $\{\delta_x\}$ to "waves" $\{e^{ikx}\}$ hints at a broader duality: it changes the very nature of the "points" that constitute our space.

Let us define a function f by its values at geometric points. In the language of distributions, we can write any function as a continuous sum of Dirac deltas:

$$f(x) = \int_{\mathbb{T}^1} f(y)\delta_y(x)\, dy$$

Here, the building blocks are points $y \in \mathbb{T}^1$. A "point" is simply a linear functional that evaluates a function at a specific location: $\mathrm{ev}_y(f) = f(y)$.

Now consider the Fourier Transform. It maps a function f on the group G to a function $\hat{f}$ on the dual group $\widehat{G}$ (the set of characters). What are the "points" of this new space? They are the irreducible representations (the frequencies).

$$\hat{f}(\chi) = \int_G f(x)\overline{\chi(x)}\, dx$$

When we transform a geometric point δ_y, it becomes a function on the dual group:

$$\widehat{\delta_y}(\chi) = \overline{\chi(y)}$$

This reveals that the Fourier Transform is an isomorphism between two different kinds of spaces:

- The *Physical Space* G, whose points are geometric locations x.
- The *Spectral Space* $\widehat{G}$, whose points are irreducible representations χ.

This is a specific instance of a grander theme in mathematics known as *Gelfand theory*, whose duality half makes a commutative C^*-algebra correspond to a topological space, its spectrum. The Fourier transform is the Gelfand transform of $L^1(G)$: it carries the algebra of functions under convolution into the algebra of functions under pointwise multiplication,

$$L^1(G) \hookrightarrow C_0(\widehat{G})$$

injectively and with dense image, but *not* onto, since not every function vanishing at infinity is the transform of an L^1 function. ($L^1(G)$ is a Banach $*$-algebra, not a C^*-algebra; the C^*-algebra whose spectrum is genuinely $\widehat{G}$ is the group C^*-algebra.) Effectively, we trade the geometry of the group for the geometry of its representation theory.

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